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Online joint-prediction of multi-forward-step battery SOC using LSTM neural networks and multiple linear regression for real-world electric vehicles

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ABSTRACT

Prediction of state of charge (SOC) is critical to the reliability and durability of battery systems in electric vehicles. The existing techniques are mostly model-based SOC estimation using experimental data, which are inefficient for learning the unpredictable battery state under complex real-world operating conditions of electric vehicles. This paper presents a novel machine-learning-enabled method to perform real-time multi-forward-step SOC prediction for battery systems using a recurrent neural network with long short-term memories (LSTM). The training results from a yearlong dataset show that the offline LSTM-based model can perform fast and accurate multi-forward-step prediction for battery SOC. Furthermore, the model has excellent practical application effects by taking into account weather and drivers' driving behaviors during real vehicular operating, and the stability, flexibility, and robustness of this method are verified by 10-fold cross-validation. In order to achieve dual control of prediction accuracy and prediction horizon, we proposed an LRLSTM-based joint-prediction strategy while using LSTM and multiple linear regression algorithms, through which an accuracy benchmark can be obtained, and the prediction steps of LSTM within acceptable accuracy can be flexibly controlled. The smooth implementation of multi-forward-step SOC prediction on real-word vehicles can eliminate drivers' mileage anxiety and safeguard vehicle operation in a big way.

1. Introduction

Battery systems have been widely used as power sources of various types of systems due to the advantages of high energy density and high safety performance, including various consumer electronics, electric vehicles (EVs), and spacecraft [1–3]. With significantly high efficiency brought about by EVs, many countries are planning to ban gasoline-powered and diesel-powered vehicles in decades. As a critical component of EVs, battery systems have been gaining increasing attention from both academia and industrial sectors because they can largely determine safety, drivability, and driving mileage of EVs. Additionally, fast-growing application markets for unmanned EVs and smart EVs are closely related to advanced battery management technologies.

Equivalent to a fuel gauge for a fuel-powered vehicle, state of charge (SOC) is a battery charge indicator for a pure EV, hybrid vehicle, or plug-in hybrid EV. SOC estimation is commonly used when people discuss the current state of a battery system, which is critical for the remaining capacity and energy of a battery system. However, SOC still cannot be obtained by direct measurement, but it can be indirectly estimated using some directly measurable battery parameters. Traditionally, SOC estimation can be realized by a nonlinear function that varies with temperature and discharge/charge currents [4], open circuit voltage-based [5] and coulomb counting techniques [6] are the two broadly used estimation tools. In recent years, more efficient methods have been developed such as Luenberger observer [7], H-infinity observer [8,9], Kalman filters(KFs) [10,11] and sliding mode observer [12], etc. However, all these methods still have two disadvantages in common: one is the significant computation sensitivity, and the other one is that insufficient parameters or inadequate models are difficult to adapt to complex actual working environments. For

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Nomenclature

IP_t	Input parameter
OP_t	Output parameter
s^t	Internal state variable
δ	Accuracy benchmark factor
H_t	Hidden state at time t
W	Weight matrix
U	"Hidden state - hidden state" matrix
ϕ	Passing coefficient

Abbreviations

ADAB	AdaBoost
BMS	Battery management system
BP	Back propagation
BPTT	Backpropagation through time
BS	Batch size
BT	Bagging Tree
DT	Decision tree
ETT	Extra Tree
EV	Electric vehicle
FNN	Feedforward neural network
GBDT	Gradient boosting decision tree
KF	Kalman filter
LR	Linear regression
LSTM	Long short-term memory
MRE	Mean relative error
PCC	Pearson correlation coefficient
PWS	Predicted window size
RF	Random forest
RMSE	Root mean square error
RNN	Recurrent neural network
RUL	Remaining useful life
SMC	Service and Management Center
SOC	State of charge
SWS	Sliding window size
TWS	Training window size

instance, some other parameters such as irreversible battery aging and bad driving behaviors, may also have large negative impacts on SOC estimation. These impacts are difficult to be accurately evaluated based on the above-mentioned methods, so as to deteriorate the SOC estimation accuracy, and prevent their application and promotion in real-world EVs.

Nowadays, great progress has been achieved in the field of artificial intelligence and machine-learning, Machine-learning algorithms are the core of artificial intelligence and its applications span various research fields such as data mining [13], computer vision [14], search engines [15], intelligent robotics [16], and autonomous vehicles [17], etc. It can be found from the summary, advancements in many of the above applications can be attributed to recurrent neural networks (RNNs). Several studies have focused on machine-learning techniques, such as feedforward neural networks (FNN) [18], back propagation (BP) neural networks [19], backpropagation through time (BPTT) [20], and truncating BPTT. In recent years, application prospects of machine-learning techniques in the field of EVs and battery management systems (BMSs) are gradually being tapped [21]. Most applications on battery systems focus on the estimation of battery SOC [22], state of power [23], state of health or remaining useful life (RUL) [24,25]. However, these above-mentioned applications are usually verified using lab-scaled experiments conducted under conventional experimental conditions, their application performances for battery systems in dynamic vehicular operation scenarios are still unpredictable.

Many classical machine-learning methods have been used to estimate battery SOC up to now. For example, Guo et al. [19] presented an improved BP neural network algorithm to prediction SOC of the lithium iron phosphate battery by significantly reducing the prediction error. Charkhgard et al. [26] applied artificial neural networks to estimate terminal voltage and fed into an adaptive extended KF to estimate battery SOC, and the root mean square error (RMSE) is as small as 2%. Meng et al. [27] used support vector machine (SVM) to estimate SOC with sliding windows, and the mean relative error (MRE) of less than 2% can be achieved. Jimenez and Lipu et al. [28,29] battery SOC estimation using a dynamic neural network and an exogenous input-based neural network, respectively. Battery voltage, current, and battery pack temperatures were taken as inputs to train the robust SOC estimation models with high accuracy. However, most of these existing machine-learning methods failed to predict multi-forward-step SOC during real-time vehicular operation, making little contribution to eliminating driver's mileage anxiety.

1.1. Challenges for battery SOC prediction in real-world EVs

At present, a large number of research methods for battery SOC estimation/prediction have been proposed, but accurate online SOC prediction in real-world EVs still have following irresistible difficulties and challenges:

1) Undoubtedly, driving conditions, weather, and drivers' driving behaviors have certain effects on battery SOC during the actual vehicular driving, which are unlikely to be obtained through experimental simulations. However, the existing researches on battery SOC estimation conducted in the laboratory environments have a strong dependence on the stability of the experimental environment, and most of the application effects on real-world vehicles under complex operating conditions are still yet to be verified.
2) Due to limitations of algorithmic capabilities, the existing methods applied to battery SOC estimation/prediction are generally single-step estimation or single-forward-step prediction. They all failed to truly eliminate drivers' mileage anxiety due to the unknown SOC fluctuation/trend over a future period of time. Therefore, a method that can accurately predict multiple-forward-step battery SOC is necessary and urgently needed for safe operation of EVs.

1.2. Contributions of the work

To address above issues, RNN with long short-term memory (LSTM) cells may be an ideal candidate for SOC prediction, which overcomes the shortcomings of gradient explosion/descent of conventional FNNs and gives RNNs the ability to sustain self-learning and predicting values for a future period of time [30,31]. This paper demonstrates a novel joint-prediction model employing the LSTM networks and multiple linear regression (LR) algorithm on real-world operational EVs, and several original contributions attempt to be made as follows:

1) A novel LSTM-based multi-forward-step SOC prediction model is presented. Compared with conventional algorithm-based [8], experiment-based or model-based methods [32], this machine-learning technique has stronger learning and memorizing abilities, and these abilities gradually increase with the accumulation of historical data [33].
2) To consider the impacts of weather and drivers' behaviors on a battery system's performance under actual operating conditions, the Weather-Vehicle-Driver analysis method is proposed, and the

proposed pre-dropout technology combined with correlation analysis can effectively prevent LSTM from overfitting.

3) In order to achieve dual control of accuracy and prediction horizon, we develop an LRLSTM-based joint-prediction model by offline training while using LR and LSTM method. Taking the one-forward-step prediction accuracy of LR as the accuracy benchmark, we can flexibly control the prediction steps of LSTM within acceptable prediction accuracy.
4) To obtain a full-climate and full-state model for the battery system in actual vehicular operation, we employ yearlong operational data of an electric taxi to train the LSTM model, which can realize long-term SOC prediction and early low-battery warning, thereby eliminating drivers' mileage anxiety.

1.3. Organization of the paper

The rest of this paper is organized as follows: Section 2 preliminarily describes the collected data and preprocessing. Section III introduces the LSTM method for multi-forward-step prediction. Section IV details the manual optimization process of hyperparameters and discussions of training results of the LSTM model. Section V shows the LRLSTM-based joint prediction strategy for battery SOC, and concludes will be drawn in Section V.

2. Data description and preprocessing

2.1. Data description

During actual vehicular operation, various battery parameters and vehicle operation parameters will fluctuate randomly with the unpredictable road conditions and driving behaviors. Big data platforms for EVs have been rapidly and massively constructed nowadays. The studied data is derived from the Service and Management Center for electric vehicles (SMC-EV) in Beijing, whose monitoring and management mechanism is sketched in Fig. 1. The SMC-EV has been upgraded to the national management platform that is monitoring millions of public EVs throughout China in real time. Besides, online information on critical vehicle components via vehicle-to-platform communication can also be obtained with a standard sampling interval of 10 s. The studied vehicle in this paper is an electric taxi assembled with a lithium-ion battery pack; the battery pack consists of 100 single cells with total energy of 29.1 kWh and rated voltage of 326 V.

2.2. Data preprocessing

To achieve a comprehensive analysis, a Weather-Vehicle-Driver analysis method is proposed for preprocessing data, by which we can simultaneously take weather factors and the driver's behaviors into consideration for SOC prediction of the studied battery system.

Through the application of the programming interface technology, weather data that may affect the vehicle's operation is obtained from the website https://www.wunderground.com, and six kinds of weather parameters are added into input analysis such as humidity, precipitation, barometric pressure, air temperature, visibility, and wind speed. Furthermore, the representative parameters of drivers' driving behaviors available from the SMC-EV platform are brake pedal stroke value, vehicle speed, and acceleration, et al. With regard to vehicle factors that may affect battery SOC, in addition to well-known battery voltage, charging/discharging current, probe temperature, RUL of the battery system is also supposed to affect battery SOC. However, acceleration and RUL are difficult and unrealistic to be calculated/monitored in real time for a real-world vehicle, and the sampling interval 10s is too large to calculate acceleration by taking the derivative of the vehicle speed. As analyzed in [31], battery RUL is negatively related to the vehicle's mileage, and acceleration value is inversely proportional to motor speed value. Therefore, to avoid errors from complex calculation/estimation, the driving mileage is used to represent the battery RUL, and the motor speed is used to represent the acceleration in this research.

By summing up the above Weather-Vehicle-Driver analysis, fifteen related parameters to battery SOC (contains the battery SOC itself) are initially determined as the input parameters to train the LSTM-based model, such as SOC, battery cell voltage, pack voltage, charging/discharging current, probe temperature, vehicle speed, brake pedal stroke value, motor speed, humidity, precipitation, barometric pressure, air temperature, visibility, wind speed, and mileage. For a typical neural network training, the probability of overfitting will increase when too many high-correlation or low-correlation parameters are trained together, which will also worsen the training effect. Not surprisingly, when we take all these parameters as inputs, the LSTM-based training process always ends at the beginning due to overfitting, no matter which of these parameters we use as the output. Therefore, a correlation analysis among all considered parameters is quite essential before modeling to eliminate the detrimental parameters. The Pearson correlation coefficients (PCCs) of these parameters are shown in Fig. 2, where the PCCs values have been done absolute value processing for



Fig. 1. The monitoring and management mechanism of SMC-EV.

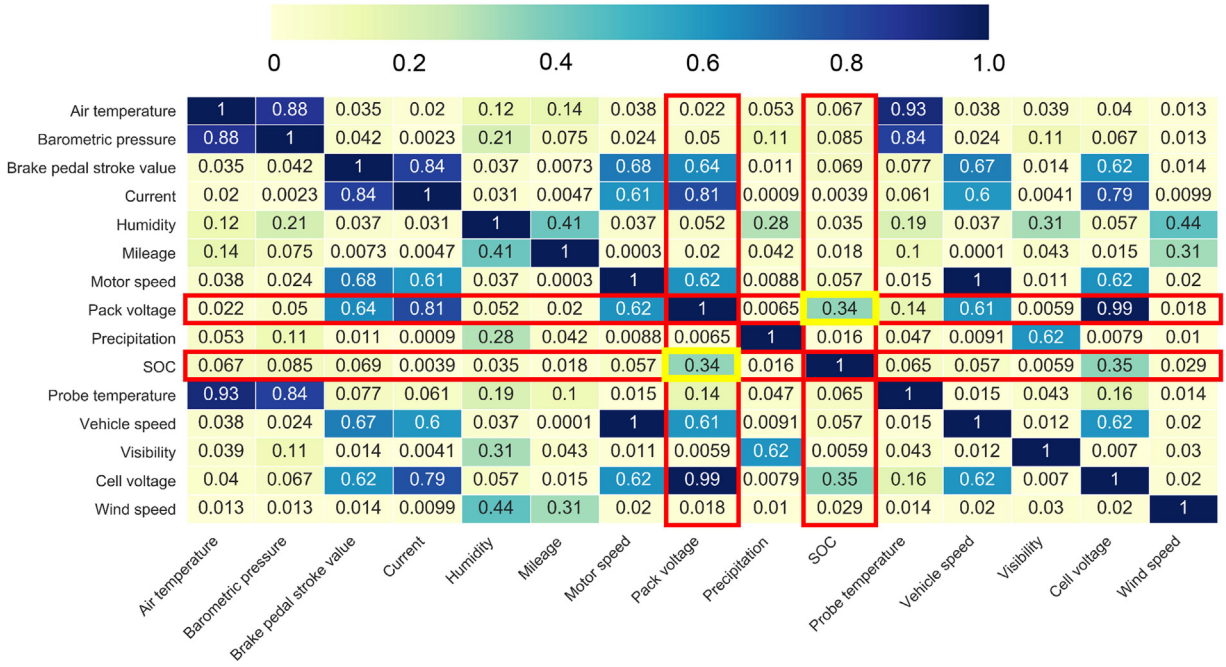


Fig. 2. The PCCs between all considered parameters.

Table 1
The relationships between correlations and the absolute values of PCCs.

Absolute value of PCCs	Correlation
0–0.2	Irrelevant or very weak correlation
0.2–0.4	Weak correlation
0.4–0.6	Moderate correlation
0.6–0.8	Strong correlation
0.8–1.0	Extremely strong correlation

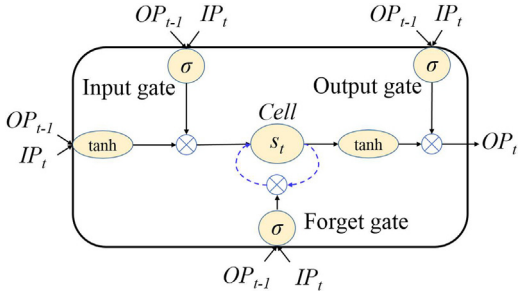


Fig. 3. The network schematics of the LSTM predictor.

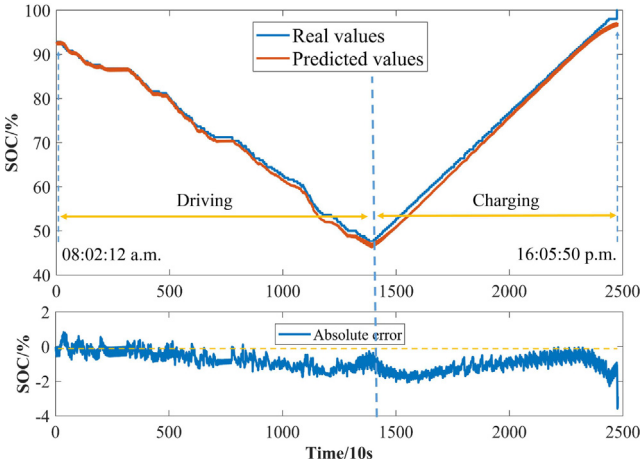


Fig. 4. The SOC prediction results and absolute errors for the studied battery system.

simplified analysis.

The correlation between two parameters according to the PCC can be illustrated in Table. 1. When we eliminate the irrelevant and extremely strong correlated parameters, the matchable parameters satisfied the training requirements are obtained as the pack voltage and SOC, which are finally determined as inputs.

3. Methodology

There are several important concerns when we apply the LSTM to predict SOC during actual vehicular operations. First, traditional deep-learning networks based on batch gradient descent or stochastic gradient descent are too slow to converge to the correct network weights, which are thus not applicable to the online prediction of battery states. Second, battery parameters of a running vehicle are diversified due to unpredictable road conditions and the driver's driving behaviors, which are highly changeable and unpredictable. Third, deep neural networks like LSTM is bound to face a risk of overfitting, which will lead to poor learning results. In order to address the above challenges, the developed LSTM predicter in this work mainly includes the following components: a many-to-one (many inputs and one output) network architecture, an Adam network optimizer [34], and a pre-dropout technology to prevent LSTM from overfitting.

RNN is arguably the most powerful neural networks with specific memory models. As a comparison, FNN delivers information straight forward (never returns nodes that have passed), while RNN passes information circularly. FNN does not have the concept of chronological order and the only considered input is the currently accepted examples, so FNN seems to suffer from short-term amnesia, which only has the memories when they were previously trained. The decision of the RNN at the t-1 time step will affect its decision at the subsequent t time step. So the RNN has two inputs from the current and previous states, respectively. This kind of feedback loop that constantly takes its last moment's output as input is the commonly said RNN's memory. The purpose of adding memory to the neural network is that the sequence

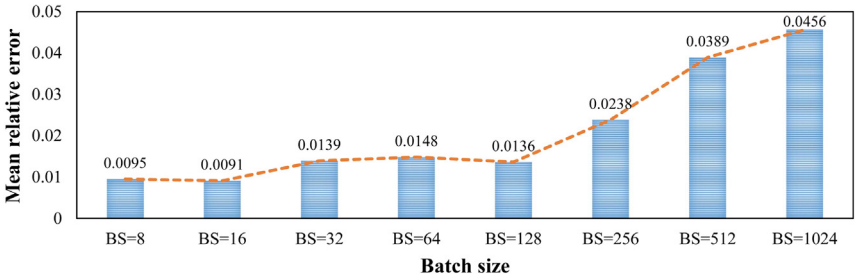


Fig. 5. MREs based on different numbers of BSs.

Table 2
The MREs of different window sizes.

Window size\Window type	PWS	TWS	SWS
6	0.0089	0.0064	0.0064
30	0.0138	0.0076	0.0076
60	0.0224	0.0043	0.0089
90	0.0239	0.0082	0.0092
120	0.0245	0.0082	0.0098
150	0.0287	0.0081	0.0101
180	0.0312	0.0085	0.0112
360	0.0358	0.0089	0.0128

itself has information, and the recurrent network can use this information to complete tasks that the FNN cannot complete.

The process of RNN passing memory forward in mathematical language can be described as:

$$H_t = \phi(W*IP_t + U*H_{t-1}) \quad (1)$$

where the hidden state at time t is H_t. It is a function of the input IP_t at the same time step, modified by a weight matrix W (as used in the FNN), plus the hidden state H_{t-1} of the previous time step multiplied by its own "hidden state - hidden state" matrix U (also called transition matrix, which is similar to Markov chain [35]). The weight matrix W is a filter that determines how much importance is given to the current input and past hidden states. The errors they produce will be returned through backpropagation and used to adjust the weights until the error can no longer be reduced. Since this feedback loop will occur at every time step of the series, each hidden state not only tracks the previous hidden state, but also includes all states before H_{t-1} within the memory capacity range.

In the early 1990s, the problem of gradient disappearance and gradient explosion has become a major obstacle to the performance of RNNs [36]. Since the layers and time steps of a deep neural network are connected to each other through multiplication, the derivatives may disappear or swell. If the gradient is unknown, the weight cannot be adjusted in the direction of reducing the error, and the network will stop learning. The problem of gradient expansion is relatively easy to solve because it can be truncated or squeezed, but the disappearing gradient may become too small for the computer to handle, which lead to this problem is more difficult to solve.

In the mid-1990s, German scholars Sepp Hochreiter and Juergen Schmidhuber [37] proposed LSTM as a RNN' variant with so-called long-term and short-term memory units that can solve the above problems. LSTMs preserve errors for reverse transfer along with time and layers. Moreover, LSTM can keep the error at a more constant level, allowing the RNN to learn over many time steps (over 1000 time steps), thereby opening the way to establishing long-term causal links. The network schematics of the LSTM predictor is shown in Fig. 3, and more details can be found in our published works [31,38]. During training, the inputs, output, and forget gates allow the LSTM to ignore or write new information into the memory cells [30,39,40]. A new input sequence IP_t is concatenated to the previous output from the cell OP_{t-1}, then the combined input is to be squashed via a tanh before passing through the input gate. The LSTM cell has an internal state variable s^t that can lag one time step. The recurrence loop is controlled by a forget gate and the output gate determines which values are allowed to be outputted from the cell OP_t.

When we use a limited dataset to optimize the LSTM learning, gradient descent is taken as an iterative process, so updating the weights using a single pass or one epoch is far from enough. When the number of epochs increases blindly, the predicted results will go from underfitting to overfitting. L1 and L2 regularization are two commonly used methods [41,42] to prevent neural networks from overfitting. However, these two methods may add extra terms to the cost function when they regularize the learning weights. Srivastava et al. [43] proposed a more efficient Dropout technique, but this technique randomly drops neurons from the networks during training, which may drop some important neurons uncontrollably during the training process. To solve the above problems, this paper proposes a pre-dropout technology to prevent LSTM from overfitting by the above correlation analysis, which can preliminary removing the parameters that are irrelevant to the target parameter and not conducive to training results. More detailed introduction about the pre-dropout technology can be found in our published work in Ref [31]. Through multiple sets of verification, the proposed pre-dropout technology can maximally preserve the neurons and connections that are beneficial to the prediction results.

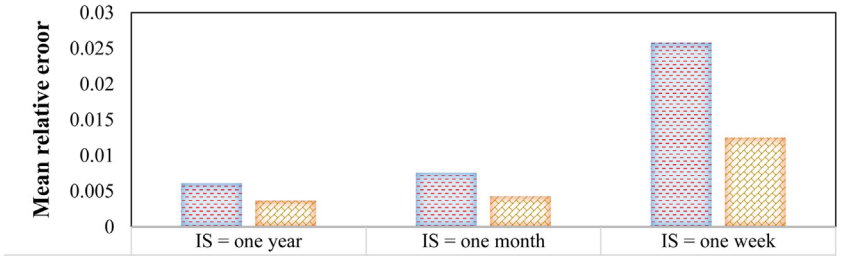


Fig. 6. MREs of different ISs between one-to-one and many-to-one model structure.

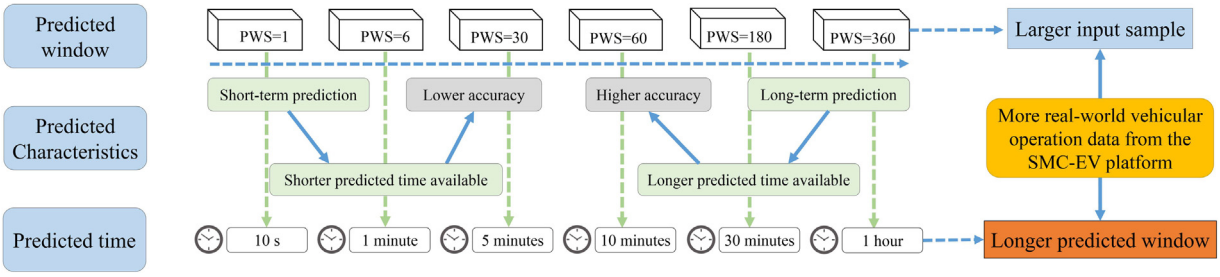


Fig. 7. The flexibility mechanism of PWSs for SOC prediction.

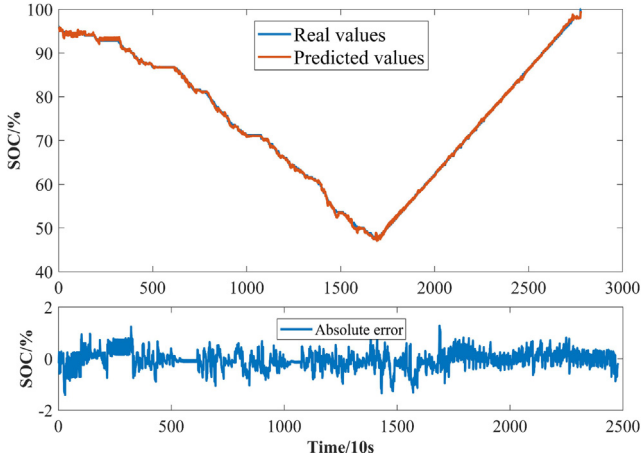


Fig. 8. The prediction results of SOC and absolute errors of the final hyperparameters.

4. Results and discussion

4.1. Selection of hyperparameters and debugging

Generally, the hyperparameters of an LSTM network need to be empirically optimized to obtain a higher model accuracy. What should be first noted during optimization is to avoid overfitting, which will lead to the model useless for prediction. A larger network is more prone to overfitting, while a larger input sample (IS) would be better for preventing overfitting. Furthermore, multiple epochs should be included in the training, which can evaluate the training performance on the testing data to determine when to stop the training. Last but not least, to learn more effective historical information, the sliding windows technique is best adopted in an allowed training configuration.

One epoch means a process that an entire data set passes forward and backward through the LSTM neural network only once. Batch size (BS) presents the number of training examples in a single batch. Besides, to let the training process learning more historical information, the sliding window technique is applied in this work. Based on empirical analysis oriented for our research object, a set of hyperparameters are initially determined as: epoch = 50, BS = 128, IS = one-month data of the year 2016, training window size (TWS) = 360 (one-hour data), predicted window size (PWS) = 6 (one-minute data), sliding window size (SWS) = 60 (10 min data). The LSTM network is constructed with one input layer, three hidden layers, one dense hidden layer with linear activation, and one output layer.

In order to compare the prediction effects of different ISs (one-year data, one-month data, and one-week data), we take one-day data on February 8, 2017, as the testing data. To ensure the stability of the well-trained model, 10-fold cross-validation is performed on the training data, where training data and validation data occupy 90% and 10% of the total IS, respectively. Furthermore, the formula of MRE to evaluate prediction results in this work can be expressed as:

$$MRE = \frac{1}{n}\sum_{i=1}^{n}\frac{|\hat{SOC}_i - SOC_i|}{SOC_i} \times 100\% \quad (2)$$

where n is the number of training or test samples, $\hat{SOC}_i$ and SOC_i are, respectively, the predicted SOC values and real SOC values (measured by the BMS) of the battery system. MRE is applied to evaluate the prediction result on the testing data, and the average MRE of ten cross-validation models is taken as the final MRE for a set of hyperparameters.

The prediction results and absolute errors based on the initial hyperparameters are indicated in Fig. 4, which indicates that the LSTM model has excellent prediction performance of battery SOC on the testing data, and the studied vehicle performs one driving and one charging on the tested day.

4.2. Determination of the LSTM network parameters

The numbers of hidden layers common in machine learning are one hidden layer, two hidden layers, and three hidden layers. By comparison calculation, the lowest average validation loss is available when there are three hidden layers. Therefore, three hidden layers are adopted in this study. For a large dataset as demonstrated in this paper, due to the memory limitations of the laptop used in this study, it is not feasible to load all the data at once, so the number of the BS requires to be adjusted within a reasonable range. In reference to other related research and literature, the BS is studied in the range of {8, 16, 32, 64, 128, 256, 512, 1024}, and the MREs of SOC prediction are calculated based on these BSs, as indicated in Fig. 5, from which we can see the

Table 3
Comparison of errors for various studies.

Method	Error	Lithium battery	Test environment
AEKF with ANN battery model [26]	RMSE < 2%	1.2 Ah	20°C (ambient)
RNN + LSTM [46]	MRE < 0.77 ~ 2.08%	2.9 Ah Panasonic LiNiCoAlO$_2$/NCA	0, 10, and 25°C (ambient)
RBFNN [47]	MRE < 5%	6 Ah LiMn$_2$O$_4$	0, 25, and 40°C (ambient)
ANN + UKF [48]	RMSE < 0.5% ~ 2.5%	2.3 Ah LiFePO$_4$	0°C to 50°C
AUKF with LSSVM battery model [27]	MRE < 2%	70 Ah Kokam	25°C (ambient)
LSTM-RNN battery model [49]	MRE < 0.6%	2.9 Ah Panasonic 18650	0, 10, and 25°C (ambient)
Fuzzy NN with genetic algorithm [50]	MRE < 0.9%	10 Ah Lyno Power	25°C (ambient)
The LSTM-based model (predictor in this paper)	MRE < 0.25%, RMSE < 0.25%	2.7Ah 18650	All-climate application

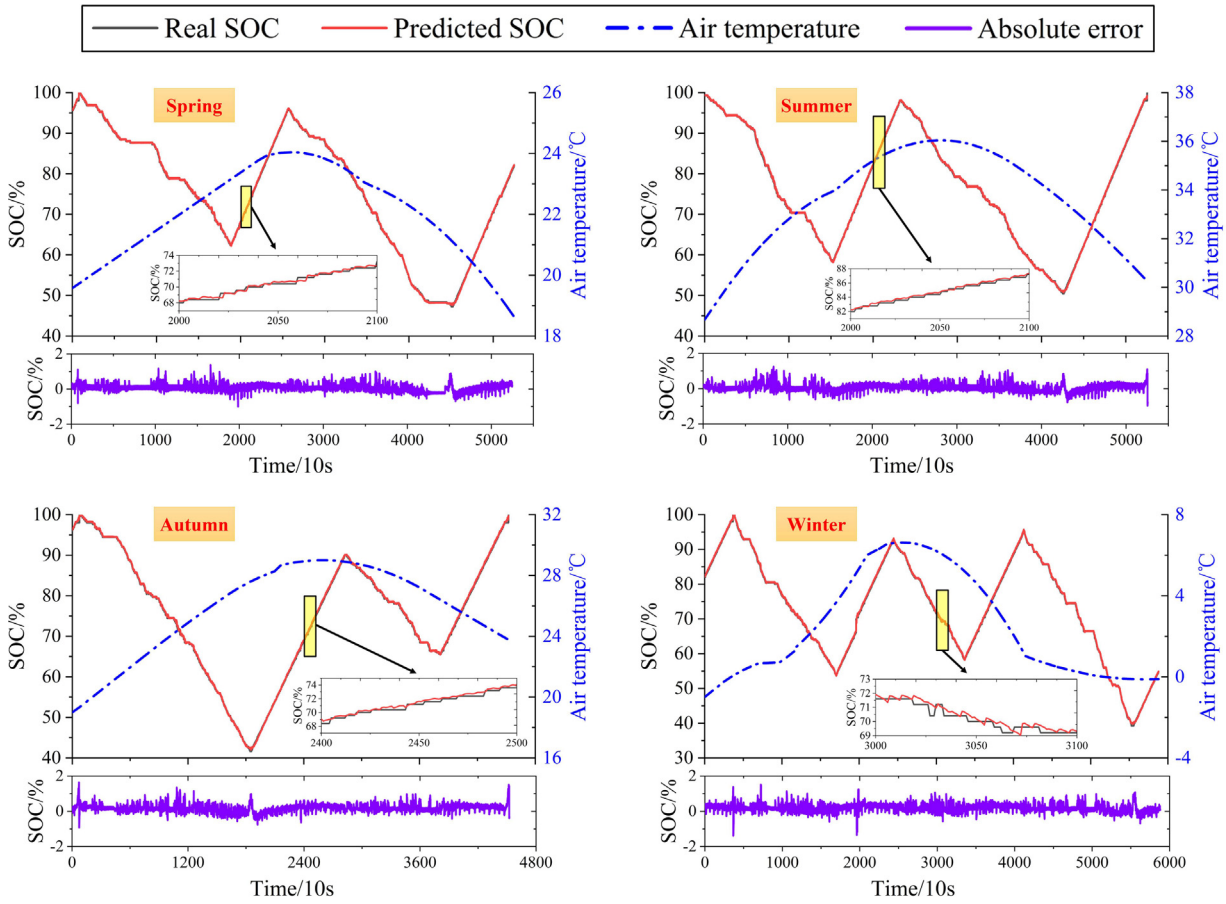


Fig. 9. The SOC curves before and after prediction in four seasons.

Table 4
The average MREs for different temperature ranges.

Temperature ranges	MREs
0~ 2°C	0.31%
2~ 4°C	0.32%
4~ 6°C	0.31%
20~ 22°C	0.29%
22~ 24°C	0.28%
24~ 26°C	0.28%
30~ 32°C	0.26%
32~ 34°C	0.25%
34~ 36°C	0.26%

prediction result with the lowest MRE appears when BS = 16.

4.3. Determination of different window sizes and ISs

Based on 10-fold cross-validation, the LSTM model is trained with different PWSs of {6, 30, 60, 90, 120, 150, 180, 360} (they respectively mean the predicted multi-forward-step SOC of 1 min, 5 min, 10 min, 15 min, 20 min, 25 min, 30 min, and 1 h). Then, based on the idea of control variables on the same hyperparameters, the MREs for different TWSs and SWSs are also calculated. The MREs of different PWSs, SWSs, and TWSs are illustrated in Table 2. Table 2 indicates that the prediction MREs gradually increase along with PWS increases, which means that a longer prediction horizon results in worse prediction accuracy. Considering the computational effort, the prediction horizon is determined as one-minute-forward data of the studied battery system, that is, PWS = 6. It can be obtained through the same analysis, the SWS and TWS are set as SWS = 6 and SWS = 60.

The prediction effects for different ISs between one-to-one model (SOC as both input and output) and many-to-one (many inputs and one output) model structures can be shown in Fig. 6, which demonstrates that the proposed many-to-one LSTM model always has better prediction results than the one-to-one model. Moreover, there is a significant decreasing trend for the MREs with IS increasing, and the MREs of IS = one-week data is much higher.

However, the research aims to predict the multi-forward-step battery SOC, so smaller PWSs don't mean better results because fewer multi-forward-step SOCs can be available, as demonstrated in Fig. 7, Therefore, how to choose PWS needs to depend on the requirements of both prediction accuracy and prediction time. To get a more accurate model with longer multi-forward-step prediction capability in the near future, the IS should be continuously enriched and updated, then the prediction horizon can continue to be extended along with hyperparameters optimization.

Summarizing the above analysis results, a set of hyperparameters can be determined as: IS = one-year data, epoch = 50, BS = 16, TWS = 60, PWS = SWS = 6. The prediction results and absolute errors of the final hyperparameters are shown in Fig. 8. Fig. 8 illustrates that the absolute errors are within a very small range, and predicted SOC values are almost coincident with the actual values, which is much better than those in Fig. 4.

4.4. Verification of superiority, stability, and robustness

This paper presents a novel LSTM-based multi-forward-step SOC prediction technology for real-word vehicles. The existing relative literature mainly focuses on estimating the battery state according to the current battery characteristics, few of them directly predict parameters for a future period such as [44,45]. The prediction results of the

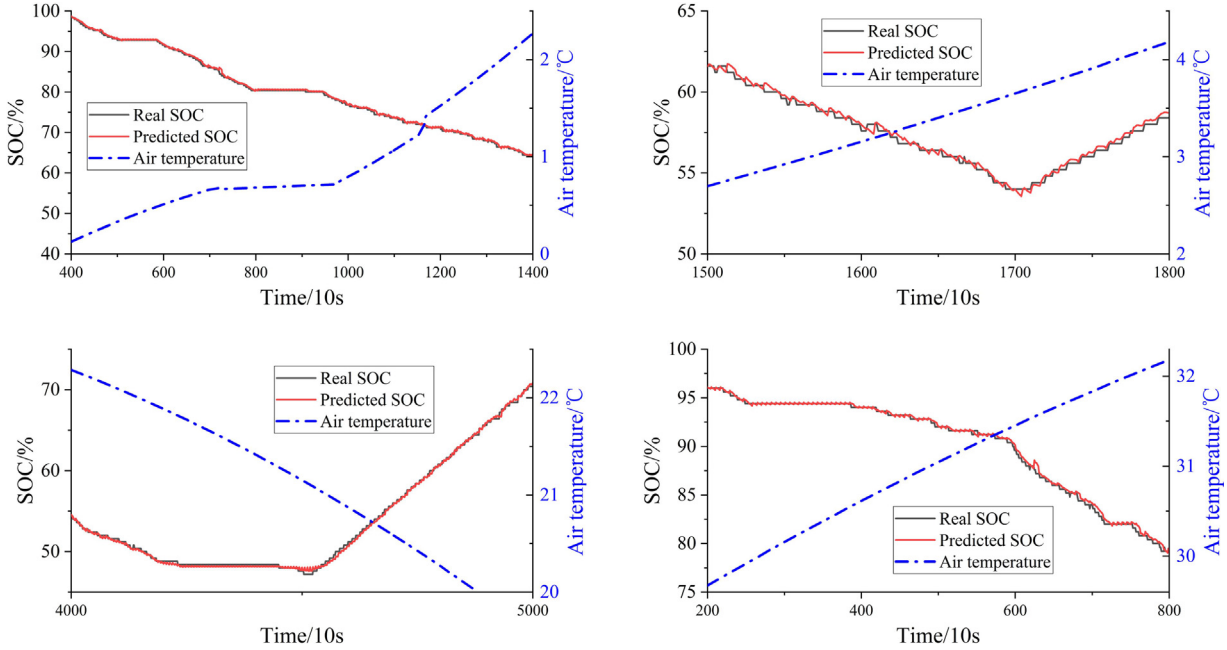


Fig. 10. The SOC curves before and after prediction in different temperature ranges.

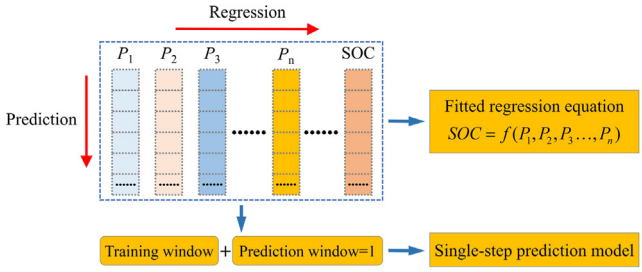


Fig. 11. The regression and prediction schematic for battery SOC.

proposed LSTM model obtained by 10-fold cross validation are compared with many other algorithms in Table 3, which indicates that the proposed method has a comparative prediction performance than other methods. More significantly, most studies in the literature were conducted in a specific laboratory environment; they are always limited to the controlled environment or charging/discharging conditions. In principle, LSTM can learn more historical characteristics for more than one year, so the proposed method can be dedicated to full-state and full-climate applications to real-world EVs without restricted by the operating environment.

To verify stability and robustness of the trained LSTM model, the SOCs of four days on the year 2017 (April 1, July 1, October 1, and January 1) are taken as the testing data to represent four seasons, and the prediction results and the absolute errors are indicated in Fig. 9. The MREs in four seasons are 0.31%, 0.26%, 0.25%, and 0.32%, respectively, which shows that the proposed LSTM model has stable prediction performance in all seasons. Fig. 9 also illustrates that the prediction effect in winter is worse than those in the other three seasons, which is inferred that the studied battery exhibits different operating characteristics under different outdoor temperatures in different seasons. Furthermore, we have also calculated the average MREs for different temperature ranges and state them in Table. 4. The partial results in four temperature ranges (0 ~ 2°C, 2 ~ 4°C, 20 ~ 22°C, and 30 ~ 32°C) are indicated in Fig. 10. Table. 4 and Fig. 10 show that the developed LSTM model has stable performance for various temperature ranges, and the model's prediction performance is slightly better under high air temperature than that under low air temperature, which is in line with the results of Fig. 9.

5. LRLSTM-Based joint prediction strategy

The SOC predictions in published studies are basically single-step predictions/estimates based on experimental data. Common methods are regression-based predictions such as decision tree (DT) [51], linear regression (LR) [52], SVM, k-Nearest Neighbor (KNN) [53], Random Forest (RF) [54], AdaBoost (ADAB) [55], Gradient Boosting Decision Tree (GBDT) [56], Bagging Tree (BT) [57], and Extra Tree (ETT) [58], etc. These methods can perform regression and single-step prediction based on different mapping directions. The schematic of regression and prediction is indicated in Fig. 11, where $P_1 \sim P_n$ represent various parameters related to battery SOC as described in Section 3.2.

The goal of regression is to map the causal relationship between the dependent and independent variables, then use mathematical models to express this specific relationship; while the purpose of prediction is to predict the unknown variables ahead of time through learning the features and correlations of historical data. Regression can be used to quantify the strength of the correlation between SOC and various parameters P_i, then can be used to evaluate parameters that are not related to SOC, while identifying which subset of Pi contains redundant information about SOC. Especially, prediction can be used to fit a prediction model to the observed parameters P_i and SOC values. When such a model is completed, for a newly added P_i value, the corresponding SOC value can be predicted using the fitted model without a matching SOC, which is the single-forward-step prediction.

To compare the prediction of regression and prediction effects using the above nine methods, we also take one-year data of the year 2016 as the IS, one-day data on February 8, 2017, as the testing data. Fig. 12 shows the regression effects on the battery SOC using above nine methods. The output is the battery SOC and the input parameters are all the considered parameters as shown in Fig. 2. It can be seen from Fig. 12 that the above nine methods generally get better regression effects in the stable charging phases than that in the complex and variable driving phases, however, they all fail to establish an accurate mapping relationship between various parameters and SOC. Many factors can contribute to this result such as an incomplete or unreasonable selection of parameters, inaccurate parameters to evaluate real vehicle operating conditions, and improper hyperparameters of the

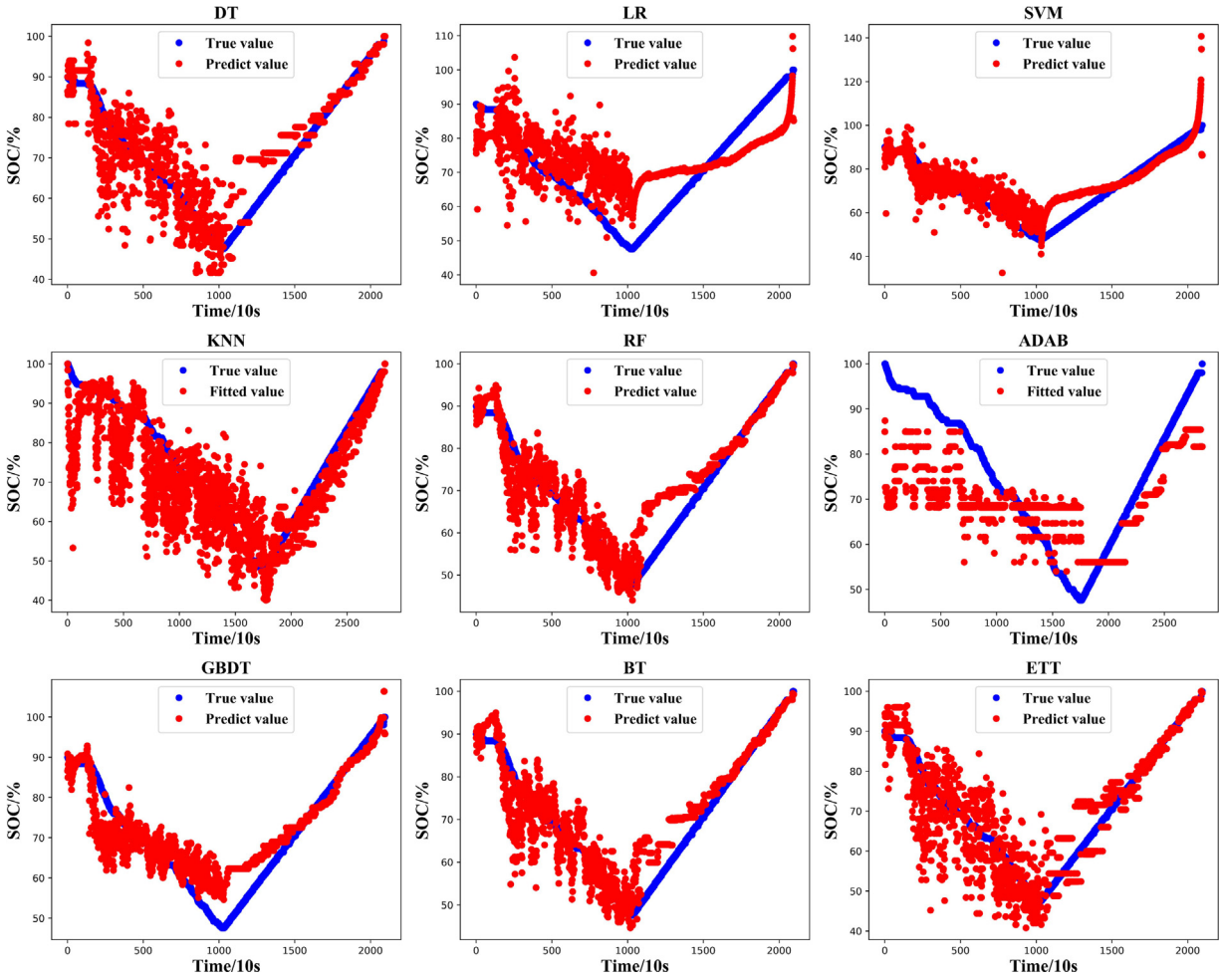


Fig. 12. SOC regression effects using different methods.

regression model, etc. Therefore, it is difficult to obtain an accurate SOC estimation equation/model through regression, which is not suitable for accurate SOC estimation in a real-world running vehicle.

In order to make a fair comparison between the above methods and the LSTM model, according to the idea of control variables, these single-step prediction algorithms are also implemented with the same input sample and training window. The SOC prediction effects and absolute errors using different methods are illustrated in Fig. 13 and Fig. 14, respectively. It is found that all the above nine algorithms can obtain good single-step prediction effects, especially the DT, LR, RF, GBDT, and BT algorithms.

Summary of one-forward-step prediction results using different methods is illustrated in Table. 5, from which we can see LR has the highest prediction accuracy, even higher than LSTM for one-forward-step SOC prediction. Combined with the above comparison of various algorithms, LR has a simpler algorithm structure for less calculation effort, which is very beneficial for us to implement online SOC prediction.

Therefore, taking the one-forward-step prediction accuracy of LR as the benchmark, we can flexibly control the prediction steps of LSTM while ensuring acceptable prediction accuracy. In order to achieve dual control of accuracy and number of prediction steps, we first establish an LRLSTM-based joint-prediction model by offline training while using LR and LSTM method. To avoid accidental errors in one-step prediction, we use the average accuracy of the predicted one step and the second step using iterative prediction of LR as the accuracy benchmark: MRE_{SS}. Then we sue the average accuracy of the first two steps of LSTM-based multi-step prediction as verification object: MRE_{MS}. When $|MRE_{SS} - MRE_{MS}| < \delta$ (δ is the accuracy benchmark factor, which id adjustable according to different demands), the predicted steps are within the acceptable range, and longer prediction steps are available. Otherwise, When $|MRE_{SS} - MRE_{MS}| \geq \delta$, the longest prediction step of LSTM is determined. Combined with the lowest acceptable prediction accuracy, the multi-forward-step SOC prediction LSTM model with adjustable steps can be obtained for fast online prediction. The LRLSTM-based multi-forward-step SOC prediction mechanism is demonstrated in Fig. 15, which can realize long-term SOC prediction and early low-battery warning.

6. Conclusion

This paper proposed an LSTM-based method for multi-forward-step SOC prediction for battery systems in real-world electric vehicles. The developed Weather-Vehicle-Driver analysis method takes into account the impact of weather and drivers' behaviors on a battery system's performance under actual operating conditions. The proposed pre-dropout technology combined with correlation analysis can not only prevent LSTM from overfitting but also effectively appoint the most suitable parameters before training. To obtain an full-climate and full-state model for the battery system in actual vehicular operation, we employ yearlong operational data of an electric taxi to train the LSTM model. The training results show that the offline LSTM-based model can perform fast and accurate multi-forward-step SOC prediction, and the stability, flexibility, and robustness of this method are verified by 10-fold cross-validation on multiple sets of actual vehicle operation data. Moreover, in order to achieve dual control of prediction accuracy and

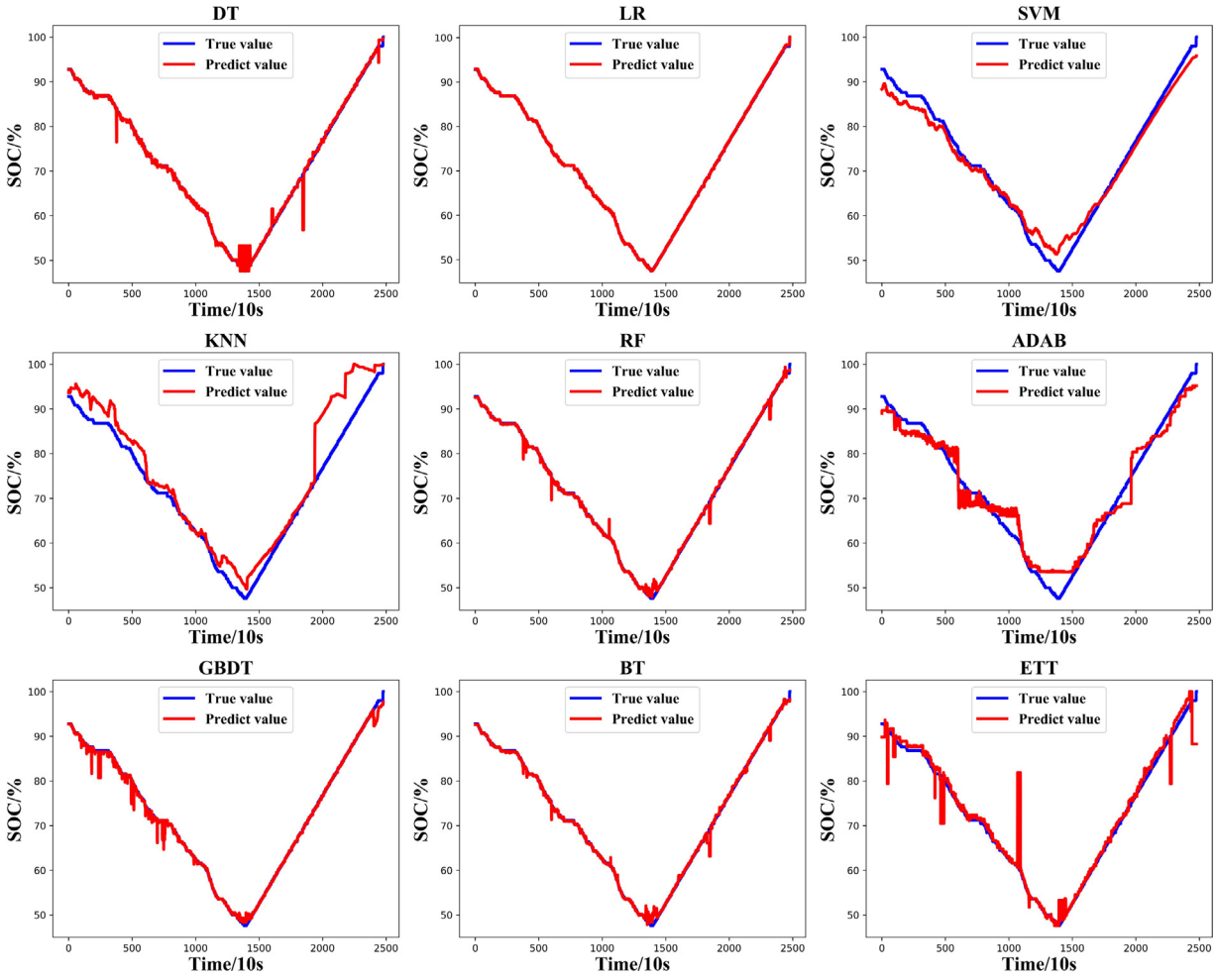


Fig. 13. SOC prediction effects using different methods.

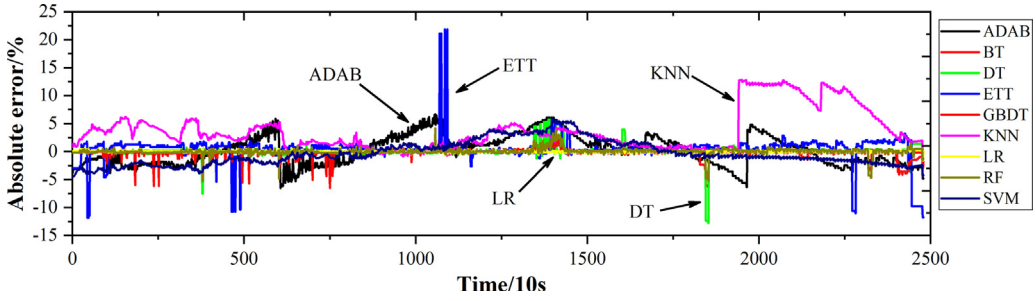


Fig. 14. Absolute errors of one-forward-step prediction using different methods.

Table 5
Comparison of one-forward-step prediction results using different methods.

Method	ADAB	BT	DT	ETT	GBDT	KNN	LR	RF	SVM	LSTM
MRE	0.033366	0.004929	0.005546	0.014188	0.004402	0.049376	0.001898	0.004534	0.027352	0.0025

prediction horizon, an LRLSTM-based joint-prediction strategy was proposed while using LSTM and multiple linear regression algorithms. Taking the one-forward-step prediction accuracy of linear regression as the accuracy benchmark, we can flexibly control the prediction steps of LSTM while ensuring acceptable prediction accuracy.

In the near future, more sets of actual vehicular data will be integrated into the input samples; especially the historical data of the same model and same batch of vehicles. This can improve the learning capability of the LSTM model for more complex vehicular operations, predicting battery SOC of more future steps, thereby greatly eliminating drivers' mileage anxiety. Furthermore, the LSTM model can be improved according to the data accumulation at any time, which makes this technology more time-sensitive, applicable, and profitable for SOC prediction. On the one hand, this method can not only perform real-time and accurate multi-forward-step SOC prediction during the actual vehicular operation, but also provide sufficient advance prediction for

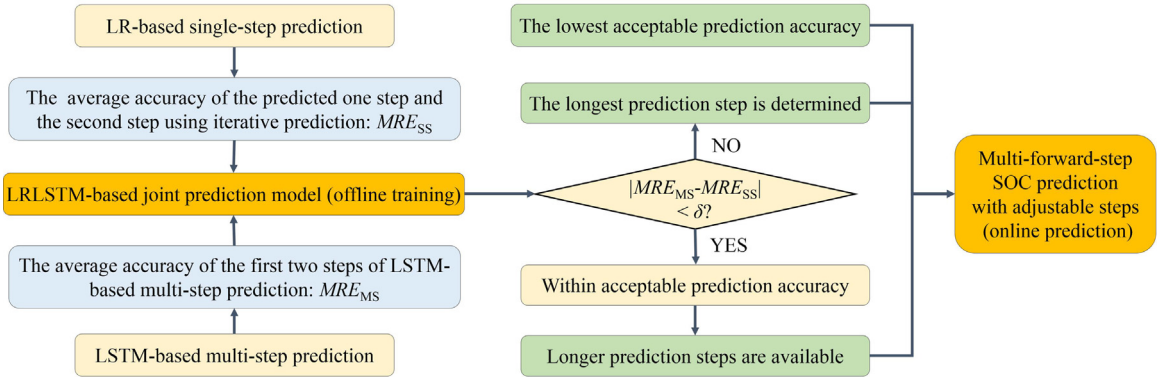


Fig. 15. LRLSTM-based multi-forward-step SOC prediction mechanism with adjustable steps.

drivers to eliminate driving mileage anxiety. On the other hand, battery anomalies/faults caused by SOC anomalies (such as low SOC, SOC jumping, etc.) can also be diagnosed in advance, thereby avoiding more serious battery faults/failures or even battery thermal runaway.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Jichao Hong: Investigation, Methodology, Writing - original draft, Writing - review & editing. **Zhenpo Wang:** Conceptualization, Funding acquisition, Project administration. **Wen Chen:** Conceptualization, Funding acquisition, Project administration. **Le-Yi Wang:** Writing - review & editing. **Changhui Qu:** Writing - review & editing.

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